

49) Show that $\neg$ and $\vee$ form a functionally complete collection of logical operators.

Soln: Let the premise stated in problem 48 hold. Wlog, let P_i be $(q_1 \wedge q_2 \wedge q_3 \wedge \dots \wedge q_n)$. By double negation law,

$$(q_1 \wedge q_2 \wedge q_3 \wedge \dots \wedge q_n) \equiv \neg(\neg(q_1 \wedge q_2 \wedge q_3 \wedge \dots \wedge q_n))$$

$$\equiv \neg(\neg q_1 \vee \neg q_2 \vee \neg q_3 \vee \dots \vee \neg q_n) \quad (\text{By 1st De Morgan's Law})$$

and Mathematical Induction, the full proof of which is in Chapter 5.

$\therefore \forall i, \exists i \wedge n$, the conjunction P_i can be written in the form shown above.

The final form has no $\wedge$ operator and hence $\neg$ and $\vee$ form a functionally complete ~~form~~ collection of logical operators $\square$.

We now present a group of exercises that involve the logical operators NAND and NOR. The proposition $p \text{ NAND } q$ is true when either p or q , or both, are false; and it is false when both p and q are true. The proposition $p \text{ NOR } q$ is true when both p and q are false, and it is false otherwise. The propositions $p \text{ NAND } q$ and $p \text{ NOR } q$ are denoted by $p \downarrow q$ and $p \uparrow q$, respectively. (The operators $\downarrow$ and $\uparrow$ are called the Sheffer stroke and the Pierce arrow after H.M. Sheffer and C.S. Pierce, respectively).

50) Construct a truth table for the logical operator NAND.

Soln:

p	q	$p \downarrow q$
T	T	F
T	F	T
F	T	T
F	F	T

51) Show that $p \downarrow q$ is logically equivalent to $\neg(p \wedge q)$.

Soln: Let $p \downarrow q$ be False. By defn. of NAND, Both p, q are True. Let $\neg(p \wedge q)$ be False. By defn. of negation, $\neg(\neg(p \wedge q))$ is True. By double-negation law, $\neg(\neg(p \wedge q)) \equiv (p \wedge q)$. $\therefore (p \wedge q)$ is True. By defn. of AND, both p, q are True. $\therefore p \downarrow q$ and $\neg(p \wedge q)$ are logically equivalent $\square$